FIXED TRANSCENDENCE DEGREE OF THE RESIDUE FIELD EXTENSION IS ETALE LOCAL ON SOURCE AND TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the notion of a property of morphisms of germs being etale local on the source-and-target; the same-chapter definition environment fixing the shape of the diagrams is delivered as a context span and is not counted as a statement.

Editorial note. The proof displays a square of local rings; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Etale local on the source-and-target: the shape of the diagrams to be considered. Same chapter, Definition definition-local-source-target-at-point, invoked at proof lines 8253-8254 and delivered as a context span. Fixes what has to be shown, namely invariance of the property under the two vertical arrows of such a diagram.
- Localisations of etale ring maps are unramified, so their residue field extensions are finite and separable. Algebra, Section algebra-section-etale, invoked at proof lines 8262-8266. Supplies the finiteness and separability of the two residue field extensions, which is what makes the transcendence degree unchanged.
- Transcendence degree is unchanged by a finite separable extension of the base field. Standard, used at proof lines 8267-8269. Gives the equality of transcendence degrees that the definition of locality demands.

Source reference. Original section 00U0: sources/stacks-chow/tags.txt, source lines 1094–1094. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Definition 0.1. Let $\mathcal{Q}$ be a property of morphisms of germs of schemes. We say $\mathcal{Q}$ is *étale local on the source-and-target* if for any commutative diagram

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

of germs with étale vertical arrows we have $\mathcal{Q}(h) \Leftrightarrow \mathcal{Q}(h')$.

Lemma 0.2. *Let $r \in \{0, 1, 2, \ldots, \infty\}$. The property of morphisms of germs*

$$\mathcal{P}_r((X, x) \to (S, s)) \Leftrightarrow \mathit{trdeg}_{\kappa(s)}\kappa(x) = r$$

is étale local on the source-and-target.

Proof. Given a diagram as in Definition 0.1 we obtain the following diagram of local homomorphisms of local rings

$$\begin{array}{ccc} \mathcal{O}_{U',u'} & \longleftarrow & \mathcal{O}_{V',v'} \\ \uparrow & & \uparrow \\ \mathcal{O}_{U,u} & \longleftarrow & \mathcal{O}_{V,v} \end{array}$$

Note that the vertical arrows are localizations of étale ring maps, in particular they are unramified (see Algebra, Section 00U0). Hence $\kappa(u')/\kappa(u)$ and $\kappa(v')/\kappa(v)$ are finite separable field extensions. Thus we have $\mathrm{trdeg}_{\kappa(v)}\kappa(u) = \mathrm{trdeg}_{\kappa(v')}\kappa(u)$ which proves the lemma. □